ENERGY SAVING USING TRAFFIC TURBINE

Submitted in partial fulfilment of Requirements for the

Award of the Degree of

BACHELOR OF TECHNOLOGY

IN

ELECTRICAL & ELECTRONICS ENGINEERING

Submitted By

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(An ISO 9001:2008 Certified institution)

(Affiliated to JNTU, Kakinada & Approved By AICTE, New Delhi)

2022- 2023

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A Project Work

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GANGURU, VIJAYAWADA - 521 139

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CERTIFICATE

This is to certify that the project work entitled **"ENERGY SAVING USING TRAFFIC TURBINE"** is a Bonafede record of project work done jointly by

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Submitted in partial fulfilment of the requirements for the award ***Degree of Bachelor of Technology*** in ***Electrical & Electronics Engineering*** to Jawaharlal Nehru Technological University, Kakinada is carried out under my guidance and supervision during the academic year 2022-2023

Mrs. KOMALI. V M.Tech	Dr . I.SAIRAM
PROJECT GUIDE	**HEAD OF DEPARTMENT**

EXTERNAL EXAMINER

ACKNOWLEDGEMENT

We are greatly indebted to **Dr. K. Ravi** principal, **DHANEKULA INSTITUTE OF ENGINEERING&TECHNOLOGY** for making us available with all the required assistance and for his support and inspiration to carry out this project in the institute.

We also take privilege to record our thank to **Dr. I. SAIRAM,** Head of the Department, ELECTRICAL & ELECTRONICS ENGINEERING whose encouragement, cooperation, and valuable support crown our success

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We are thankful to the teaching and non-teaching staff of EEE department for their direct as well as indirect help in our project.

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INSTITUTE

VISION

Pioneering Professional Education through Quality

MISSION

- Providing Quality Education through State-of-art, infrastructure, laboratories and committed staff.
- Molding Students as proficient, competent and socially responsible engineering personnel with ingenious intellect.
- Involving faculty members and students in research and development works for betterment of society.

DEPARTMENT

VISION

Emerge as Quality Human Resource Provider for Industry and Society in the field of Electrical & Electronics Engineering.

MISSION

- Providing Quality Education through State-of-art resources.
- To develop innovative, proficient, Electrical Engineers.
- Promoting Ethical and moral values among the students so as to make them responsible professionals for the society.

PROGRAM OUTCOMES (POs)

Engineering graduates will be able to:

1. Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.

2. Problem Analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching sustained conclusions using first principles of mathematics, natural sciences, and engineering sciences.

3. Design/Development of Solutions: Design solutions for complex engineering problems and design system components or process that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal and environmental considerations.

4. Conduct Investigations of Complex Problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

5. Modern Tool Usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

6. The Engineer and Society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

7. Environment and Sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of and need for sustainable development.

8. Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

9. Individual and Team Work: Function effectively as an individual, and as a member or a leader in diverse teams, and in multidisciplinary settings.

10. Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

11. Project Management and Finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multi-disciplinary environments.

12. Life- Long Learning: Recognize the need for and have the preparation and ability to engage in independent and life- long learning in broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOs)

PSO1: Ability to design solutions for identified problems by using latest engineering tools like MATLAB, Simulink, PSPICE, PLC etc.

PSO 2: Able to design and develop the Green Electrical systems.

Quality Index Assessment

Academic Year: 2022-2023

Title of the Project: ENERGY SAVING USING TRAFFICTURBINE

Parameter	Product/ Research	Working Model /Analysis/ Application	Review / Others	Consideration for Cost Effectiveness	Consideration for Environment	Safety Consideration	Standards	Followed Ethics
Level	3	3	1	High – 3 Moderate – 2 Low-1	High – 3 Moderate-2 Low-1	High – 3 Moderate – 2 Low-1	High – 3 Moderate – 2 Low-1	High – 3 Moderate – 2 Low-1
Weightage	60%			6%	10%	10%	4%	10%
Project Level	a =2			b =2	c =3	d =3	e =2	f =3

Quality Index of Project = (a * 0.6) + (b * 0.06) + (c * 0.1) + (d * 0.1) + (e * 0.04) + (f * 0.1)
= (2*0.6) + (2*0.06) + (3*0.1) + (3*0.1) + (2*0.04) + (3*0.1) =2.22

Signature of Guide Signature of Project Coordinator Signature HOD

ABSTRACT

ENERGY SAVING USING TRAFFIC TURBINES

The main objective of our project is to design a wind turbine to recapture wind energy from vehicles on the highway. Wind energy is considered the fastest growing clean energy source however; it is limited by variable natural wind. Every year wind energy generation is increasing. Highways can provide a considerable amount of wind to drive a turbine due to high vehicle traffic. This kinetic energy is unused. Research on wind patterns was used to determine the average velocity of the wind created by oncoming vehicles is 60 – 90 mph. The wind turbines are designed to be placed on the medians therefore fluid flow from both sides of the highway will be considered in the design. Moreover 20-30 mph is enough to create energy from the Traffic Turbine. The design of the turbines consists of blades, Buck-Boost convertor, collars, bearings, a shaft, gears and a generator. Additionally, since the wind source will fluctuate, a storage system for the power generated was designed to distribute and maintain a constant source of power. This power will be distributed to streetlights. Ideally, the turbine can be used globally as an unlimited power source for streetlights and other government applications.

KEYWORDS: Traffic turbine, Buck-Boost convertor, Energy storage system.

Mapping with PO/PSOs along with justification:

Title of Project	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11	PO 12	PSO 1	PSO 2
ENERGY SAVING USING TRAFFIC TURBINES	-	3	3	2	2	3	3	2	3	3	3	3	2	3

Justifications :(Project Outcomes with respect to Mapped PO`s & PSO`s):

- Our project maps strongly to po2 because in our project we just know the how to covert wind energy into electrical energy and it does not involve the first principles of mathematics, natural sciences.
- Our project maps strongly to po3 because our project solution is to utilize the unused flow of wind caused due to the vehicles moving in National highways.
- Our project maps averagely to po4 because it uses the knowledge and methods according to research for output.
- Our project maps average to po5 because it involves modern tools
- Our project maps strongly to po6 because it partly related to the responsibilities of engineering practices.
- Our project maps strongly to po7 as our project is helpful to the society and it does not have any impact on environmental issues.
- Our project maps averagely to po8 i.e., ethics and values
- Our project maps strongly to po9 because we function effectively as individual or as a team.
- Our project maps strongly to po10 as we communicate effectively about our project • Our project maps strongly to po11 i.e., project management and finance
- Our project maps strongly to po12 i.e., lifelong learning.
- Our project maps averagely to pso1 as it uses modern tool usage for software.
    - Our project maps strongly to pso2 because it is able to design green electrical system.

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CHAPTER-1

INTRODUCTION

Wind power is the use of air flow through wind turbines to mechanically power generators for electric power. Wind power, as an alternative to burning fossil fuels, is plentiful, renewable, widely distributed, clean, produces no greenhouse gas emissions during operation, consumes no water, and uses little land. The net effects on the environment are far less problematic than those of non-renewable power sources. Wind power is the use of air flow through wind turbines to mechanically power generators for electric power. Wind power, as an alternative to burning fossil fuels, is plentiful, renewable, widely distributed, clean, produces no greenhouse gas emissions during operation, consumes no water, and uses little land. The net effects on the environment are far less problematic than those of non-renewable power sources. Wind power gives variable power which is very consistent from year to year but which has significant variation over shorter time scales. It is therefore used in conjunction with other electric power sources to give a reliable supply. As the proportion of wind power in a region increases, a need to upgrade the grid, and a lowered ability to supplant conventional production.

WIND TURBINE

1.1 Wind Turbines:

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Figure 1.1 wind energy diagram

1.2 Types of wind turbines:

Wind turbines can be separated into two basic types determined by which way the turbine spins. Wind turbines that rotate around a horizontal axis are more common (like a wind mill), while vertical axis wind turbines are less frequently used (Savonius and Darrius are the most common in the group).

1.2.1 Horizontal axis wind turbine (HAWT):

The horizontal wind turbine is a turbine in which the axis of the rotor's rotation is parallel to the wind stream and the ground. Most HAWTs today are two- or three-bladed, though some may have fewer or more blades. The HAWT works when the wind passes over both surfaces of the air foil shaped blade but passes more rapidly at the upper side of the blade, thus, creating a lower pressure area above the air foil. The difference in the pressures of the top and bottom surfaces results in an aerodynamic lift. The blades of the wind turbine are constrained to move in a plane with a hub at its centre, thus, the lift force causes rotation about the hub.



Figure 1.2.1(a) Horizontal axis wind turbine

Horizontal axis wind turbine

1.2.2 Vertical axis wind turbine (VAWT):

The vertical axis wind turbine is an old technology, dating back to almost 4,000 years ago. Unlike the HAWT, the rotor of the VAWT rotates vertically around its axis instead of horizontally. Though it is not as efficient as a HAWT, it does offer benefits in low wind situations wherein HAWTs have a hard time operating. It tends to be easier and safer to build, and it can be mounted close to the ground and handle turbulence better than the HAWT.



Figure 1.2.2vertical axis wind turbine

1.3 WORKING OF BLADES

1.3.1 Working of horizontal axis wind turbine:

The wind passes over both surfaces of the air foil shaped blade but passes more rapidly over the longer (upper) side of the air foil, thus creating a lower-pressure area above the air foil. The pressure differential between top and bottom surfaces results in aerodynamic lift. In an aircraft wing, this force causes the air foil to rise, lifting the aircraft off the ground. Since the blades of a wind turbine are constrained to move in a plane with the hub as its centre, the lift force causes rotation about the hub. In addition to the lift force, a drag force perpendicular to the lift force impedes rotor rotation. A prime objective in wind turbine design is for the blade to have a relatively high lift-to-drag ratio. This ratio can be varied along the length of the blade to optimize the turbine's energy output at various wind speeds.

1.3.2 Working of vertical axis wind turbine:

A wind turbine secures air into a hub, which them turns into a generator. The air that passes through the blades of the wind turbine is spun into the generator through rotational momentum. The position of the blades is different in the VAWT. On this model, the base of the tower holds the generator, and the blades then wrap themselves around the shaft. People use the VAWT because they can be placed

closer to the ground, which makes them acceptable and effective for use at a residential location. With the vertical axis wind turbine, the rotor shaft is arranged in a vertical pattern. The VAWT are easier and more affordable to maintain than horizontal units. One complains that some users have with the VAWT is that is creates less wind energy, which may cause several different noises to be heart. Turbulent air flow is also a possibility that can shorten the life of the system.

Working of vertical axis wind turbine Preference of VAWT with HAWT:

1. VAWT has high mechanical stability because it is near to ground level.
2. It accepts wind from any angle.
3. It does not require complex mechanism (yaw, pitch control).
4. The Initial cost is low as compared to horizontal axis wind turbine.
5. Efficiency is high when compared to horizontal axis wind turbine.

1.3.3 Working of wind turbine:

Wind turbines operate on a simple principle. The energy in the wind turns two or three propellerlike blades around a rotor. The rotor is connected to the main shaft, which spins a generator to create electricity. Click on the image to see an animation of wind at work. Simply stated, a wind turbine works the opposite of a fan. Instead of using electricity to make wind, like a fan, wind turbines use wind to make electricity. The wind turns the blades, which spin a shaft, which connects to a generator and makes electricity.

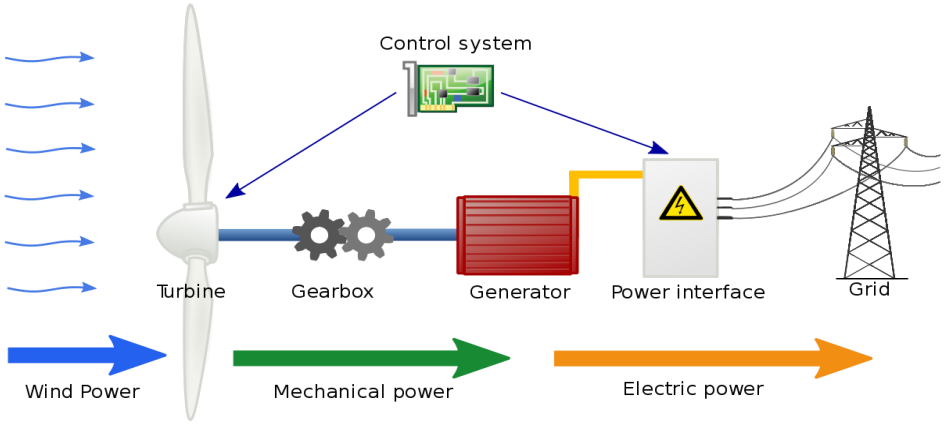


Figure 1.3.3 working of wind turbine

CHAPTER - 2

OBJECTIVE

2.1 MOTIVATION AND OBJECTIVE

The motivation for designing a highway wind turbine is to contribute towards the global trend in wind energy production in a feasible way. Wind turbines are traditionally employed in rural areas, the goal of this project is to design a wind turbine that can be used in cities. In particular, the turbines will use the wind draft created by vehicles on the highway to generate electricity. The idea is to offset the amount of pollution created by burning fossil fuels by introducing a potential source of clean energy.

2.2 CONCEPTUAL DESIGN (DESIGN ALTERNATIVES)

There are several ways to approach this design problem. In literature surveys, we discovered different features of wind turbines which were appealing for different reasons. For example, the gear turbines in China were very inexpensive and the modular sections could easily be snapped together to form a bigger system. That design did not seem as environmentally friendly as the designs with larger propellers. Other designs include turbines built into highway dividers or on overhead poles as seen in the design by the Arizona State Student Joe (last name not provided) (Joe, 2007). Joe calculated that with cars moving at 70 mph, 9,600 kilowatts of electricity could be produced per year using his design.

CHAPTER -3

COMPONENETS DESCRIPTION

3.1 ARDUINO

3.1.1 What is Arduino?

Arduino is an open-source electronics platform based on easy-to-use hardware and software. Arduino boards are able to read inputs - light on a sensor, a finger on a button, or a Twitter message - and turn it into an output - activating a motor, turning on an LED, publishing something online. You can tell your board what to do by sending a set of instructions to the microcontroller on the board. To do so you use the Arduino programming language (based on Wiring), and the Arduino Software (IDE), based on Processing.

Arduino was born at the Ivrea Interaction Design Institute as an easy tool for fast prototyping, aimed at students without a background in electronics and programming. As soon as it reached a wider community, the Arduino board started changing to adapt to new needs and challenges, differentiating its offer from simple 8-bit boards to products for IoT applications, wearable, 3D printing, and embedded environments. All Arduino boards are completely open-source, empowering users to build them independently and eventually adapt them to their needs. The software, too, is open-source, and it is growing through the contributions of users worldwide.

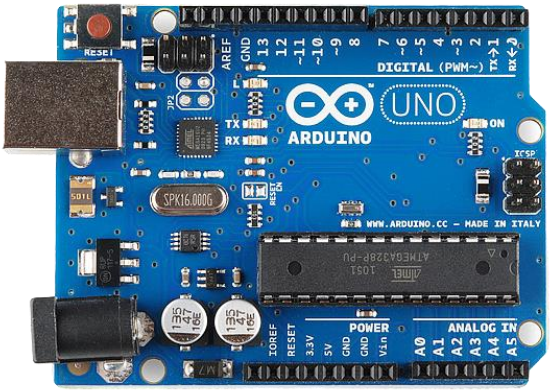


FIG.3.1 Arduino Microcontroller

3.1.2 Why Arduino?

Thanks to its simple and accessible user experience, Arduino has been used in thousands of different projects and applications. The Arduino software is easy-to-use for beginners, yet flexible enough for advanced users. It runs on Mac, Windows, and Linux. Teachers and students use it to build low-cost scientific instruments, to prove chemistry and physics principles, or to get started with programming and robotics. Designers and architects build interactive prototypes, musicians and artists

use it for installations and to experiment with new musical instruments. Makers, of course, use it to build many of the projects exhibited at the Maker Faire, for example. Arduino is a key tool to learn new things. Anyone - children, hobbyists, artists, programmers - can start tinkering just following the step-by-step instructions of a kit, or sharing ideas online with other members of the Arduino community. There are many other microcontrollers and microcontroller platforms available for physical computing.

- Cross-platform - The Arduino Software (IDE) runs on Windows, Macintosh OSX, and Linux operating systems. Most microcontroller systems are limited to Windows.
- Simple, clear programming environment - The Arduino Software (IDE) is easy-to-use for beginners, yet flexible enough for advanced users to take advantage of as well. For teachers, it's conveniently based on the Processing programming environment, so students learning to program in that environment will be familiar with how the Arduino IDE works.
- Open source and extensible software - The Arduino software is published as open source tools, available for extension by experienced programmers. The language can be expanded through C++ libraries, and people wanting to understand the technical details can make the leap from Arduino to the AVR C programming language on which it's based. Similarly, you can add AVR-C code directly into your Arduino programs if you want to.
- Open source and extensible hardware - The plans of the Arduino boards are published under a Creative Commons license, so experienced circuit designers can make their own version of the module, extending it and improving it. Even relatively inexperienced users can build the breadboard version of the module in order to understand how it works and save money.

3.2 DC GENERATOR:

Electrical generators are standalone machines that provide electricity when power from the local grid is unavailable. These generators supply backup power to businesses and homes during power outages. Generators do not create electrical energy, but they convert mechanical or chemical energy into electrical energy. Based on the output, generators are classified into two types as AC generators and DC generators. In this article, we will be discussing DC generators in detail. You can check out our article on the AC generator to understand its working principle, construction and more. According to Faraday's law of electromagnetic induction, we know that when a current-carrying conductor is placed in a varying magnetic field, an emf is induced in the conductor. According to Fleming's right-hand rule, the direction of the induced current changes whenever the direction of motion of the conductor changes. Let us consider an armature rotating clockwise and a conductor at the left moving upwards. When the armature completes a half rotation, the direction of the motion of the conductor will be reversed downward. Hence,



Fig.3.2 12-36v Dc Generator

the direction of the current in every armature will be alternating. But with a split ring commutator, connections of the armature conductors get reversed when a current reversal occurs. Therefore, we get a unidirectional current at the terminals.

3.3 NICKEL–CADMIUM BATTERY

The nickel–cadmium battery (Ni–Cd battery or NiCad battery) is a type of rechargeable battery using nickel oxide hydroxide and metallic cadmium as electrodes. The abbreviation *Ni-Cd* is derived from the chemical symbols of nickel (Ni) and cadmium (Cd): the abbreviation *NiCad* is a registered trademark of saft corporation, although this brand name is commonly used to describe all Ni–Cd batteries.

Wet-cell nickel-cadmium batteries were invented in 1899. A Ni-Cd battery has a terminal voltage during discharge of around 1.2 volts which decreases little until nearly the end of discharge. The maximum electromotive force offered by a Ni-Cd cell is 1.3 V. Ni-Cd batteries are made in a wide range of sizes and capacities, from portable sealed types interchangeable with carbon-zinc dry cells, to large ventilated cells used for standby power and motive power. Compared with other types of rechargeable cells they offer good cycle life and performance at low temperatures with a fair capacity but their significant advantage is the ability to deliver practically their full rated capacity at high discharge rates (discharging in one hour or less). However, the materials are more costly than that of the Lead acid battery, and the cells have high self-discharge rates.

Sealed Ni-Cd cells were at one time widely used in portable power tools, photography equipment, flash lights emergency lighting, hobby Rs, and portable electronic devices. The superior capacity of nickel metal hydride battery, and recent lower cost, has largely supplanted Ni-Cd use. Further, the environmental impact of the disposal of the toxic metal cadmium has contributed considerably to the reduction in their use. Within the European Union, Ni-Cd batteries can now only be supplied for replacement purposes or for certain types of new equipment such as medical devices.



FIG.3.3 Nickel Cadmium Batteries

3.4 LED

A light-emitting diode (LED) is a semiconductor device that emits light when current flows through it. Electrons in the semiconductor recombine with electron holes, releasing energy in the form of photons. The colour of the light (corresponding to the energy of the photons) is determined by the energy required for electrons to cross the band gap of the semiconductor.[5] White light is obtained by using multiple semiconductors or a layer of light-emitting phosphor on the semiconductor device.

Appearing as practical electronic components in 1962, the earliest LEDs emitted low-intensity infrared (IR) light.[7] Infrared LEDs are used in remote-control circuits, such as those used with a wide variety of consumer electronics. The first visible-light LEDs were of low intensity and limited to red. Early LEDs were often used as indicator lamps, replacing small incandescent bulbs, and in seven-segment displays. Later developments produced LEDs available in visible, ultraviolet (UV), and infrared wavelengths, with high, low, or intermediate light output, for instance white LEDs suitable for room and outdoor area lighting. LEDs have also given rise to new types of displays and sensors, while their high switching rates are useful in advanced communications technology with applications as diverse as aviation lighting, fairy lights, automotive headlamps, advertising, general lighting, traffic signals, camera flashes, lighted wallpaper, horticultural grow lights, and medical devices.

LEDs have many advantages over incandescent light sources, including lower power consumption, longer lifetime, improved physical robustness, smaller size, and faster switching. In exchange for these generally favourable attributes, disadvantages of LEDs include electrical limitations to low voltage and generally to DC (not AC) power, inability to provide steady illumination from a pulsing DC or an AC electrical supply source, and lesser maximum operating temperature and storage temperature.

In contrast to LEDs, incandescent lamps can be made to intrinsically run at virtually any supply voltage, can utilize either AC or DC current interchangeably, and will provide steady illumination when powered by AC or pulsing DC even at a frequency as low as 50 Hz. LEDs usually need electronic support components to function, while an incandescent bulb can and usually does operate directly from an unregulated DC or AC power source.

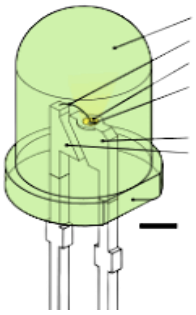


FIG.3.4 **Led**

3.5 LDR

LDR (Light Dependent Resistor) as the name states is a special type of resistor that works on the photoconductivity principle means that resistance changes according to the intensity of light. Its resistance decreases with an increase in the intensity of light. It is often used as a light sensor, light meter, Automatic street light, and in areas where we need to have light sensitivity. It is also called a Light Sensor.

The Light-dependent resistors made with photosensitive semiconductor materials like Cadmium Sulphides (CDs), lead sulphide, lead selenide, indium antimonide, or cadmium selenide and they are placed in a Zig-Zag shape as you can see in the pic below. Two metal contacts are placed on both ends of the Zig-Zag shape these metal contacts help in creating a connection with the LDRs.

It works on the principle of photoconductivity whenever the light falls on its photoconductive material, it absorbs its energy and the electrons of that photoconductive material in the valence band get excited and go to the conduction band and thus increasing the conductivity as per the increase in light intensity.

Also, the energy in incident light should be greater than the bandgap gap energy so that the electrons from the valence band got excited and go to the conduction band.

The LDR has the highest resistance in dark around 1012 Ohm and this resistance decreases with the increase in Light.

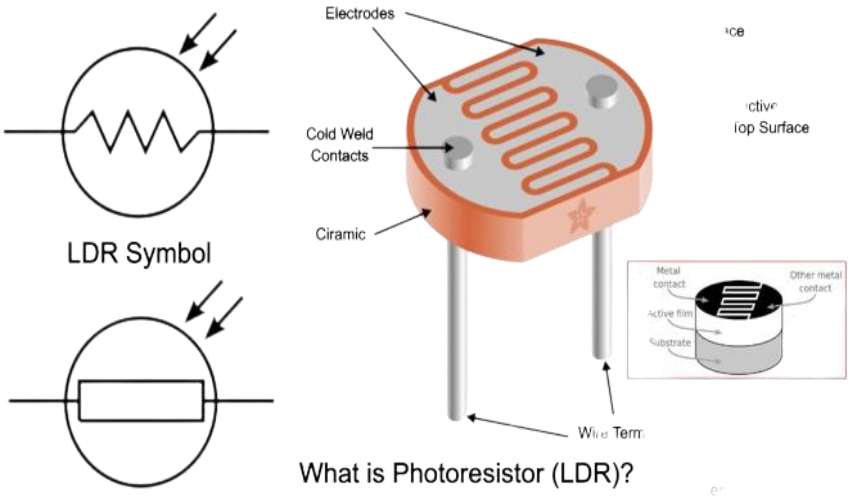


FIG.3.5 LDR

LDR Applications

- The photoresistor is generally used in detecting the presence and intensity of light
- Used in automatic lights that switch on and off according to light
- Simple Smoke Detector Alarm, Clock with automatic light
- Optical circuit design
- Photo proximity switch
- Laser-based security systems
- Solar Street Lamps
- Camera light meters
- Clock radios

3.6 RELAYS:

This is a small and easy to use 1 channel relay board that operates on 12V. Use it to control one 240V power appliance directly from Arduino, Raspberry Pi, and other microcontrollers or low voltage circuits. Perfect for switching 240V appliances lights, fans, etc, and even high power motors at lower voltages.

The board uses a high-quality relay, which can handle a maximum of 10A/250 V AC or 15A/125V AC. Each relay has all three connections Common, Normally Open, Normally Closed brought out to 3 pin screw terminals which make it easy to make and remove connections. The board has a power indication and a relay status LED to ease debugging. The board can accept control inputs within a wide range of voltages from 4V to 12V. Power input and relay control signals are brought to header pins on the board. Hence, the board can be easily interfaced with our development boards using our female-to-female jumper wires.



FIG.3.6 RELAY

Features:

i. Brand new and high quality.

ii. 1-Channel Relay interface board,

iii. Equipped with high-current relay: 15A @ 125V AC or 10A @ 250V AC

iv. Standard interface that can be controlled directly by microcontroller (Arduino, 8051, AVR, PIC)

3.7 PINION GEARS

- A pinion is a round gear usually the smaller of two meshed gears—used in several applications, including drivetrain and rack and pinion systems. Drivetrains usually feature a gear known as the pinion, which may vary in different systems.
- The typically smaller gear in a gear drive train (although in the first commercially successful steam locomotive the Salamanca the pinion was rather large). In many cases, such as remote controlled toys, the pinion is also the drive gear for a reduction in speed, since electric motors operate at higher speed and lower torque than desirable at the wheels. However, the reverse is true in watches, where gear trains commence with a high-torque, low-speed spring and terminate in the fast-and-weak escapement.

- The smaller gear that drives in a 90-degree angle towards a crown gear in a differential drive.
- The small front sprocket on a chain driven motorcycle.
- The clutch bell gear when paired with a centrifugal clutch, in radio-controlled cars with an engine



FIG.3.7 PINIONS

CHAPTER- 4

BLOCK DIAGRAM

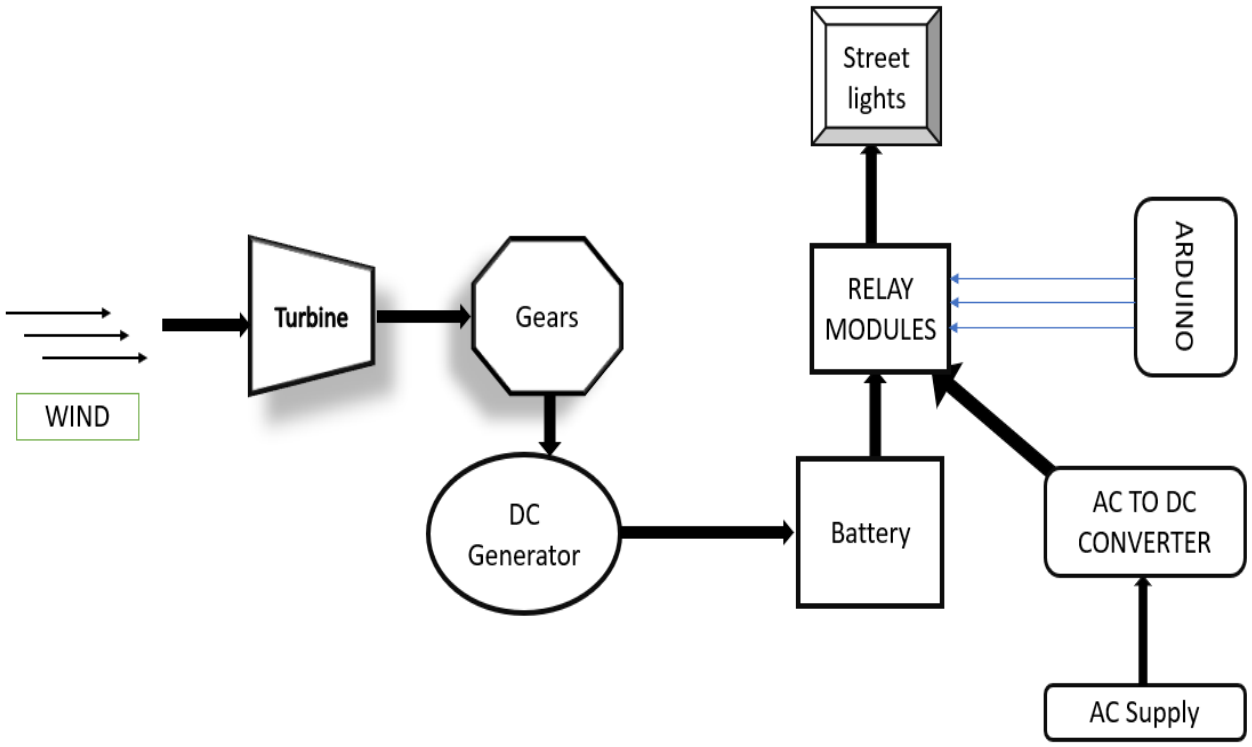


FIG.4.1 BLOCK DIAGRAM

The above fig shows block diagram representation shows a vertical axis wind turbine to utilize the wind produced by moving vehicles to generate electricity. These turbines will be placed along roadways that have high volume of fast-moving traffic. The electricity generated will then be the stored in batteries from buck boost converter. Since the electricity produced will be direct current (DC) it must be used for lighting the street lamps or any of the man ways we use electricity today. while the charge of the battery is not sufficient the relay circuit is activated and the system is fed with supply mains [AC] This means that the ac current from the mains must be pasted through a converter first before it is used.

CHAPTER- 5

5.1 TEST RESULT

The project aim is to provide uninterrupted power supply to rural areas and street lightening and to find solution to save power. In this project the first thing to do is to prepare the inputs and outputs of the system to control the light of street. The prototype is as shown in the figure below. It has been implemented and works as expected and will prove to be very useful and will fulfil all the present constraints if it is implemented on large scale.





FIG.5.1 OUTPUT VOLTAGE **FIG.5.2 OUTPUT CURRENT**

The above figure shows the output voltage and current energy generated from the turbine when it meets certain condition like minimum rpm, wind flow when it is placed in medians of highways.

5.2 Wind Turbine Results:

Wind is not constant but it can operate in day and night time also, power is generated with respect to the flow of wind through the turbine blades. Parameters of generator is given below table.

S.No	Parameters	Ratings
1.	Voltage	2V
2.	No load current	60mA
3.	Load current	130mA
4.	Torque (T)	1.5Kg
5.	Weight	125gms
6.	RPM	320RPM

CHAPTER- 6

6.1 PROPOSED SYSTEM

PROJECT FABRICATION

CUTTING AND SOFTENING

We have taken a ¾ inch square pipe to create the base and the body to withstand the wind speed

By using a cutting machine have cut the square pipe into required sizes based on the size of the turbine shaft

.



FIG.6.1 CUTTING

WELDING

By using the arc welding process, we molded the square pipe into our desired shape



FIG.6.2 WELDING

ASSEMBLING

Project after assembling.



FIG.6.3 ASSEMBLING

CHAPTER -7

7.1 RESULTS AND DISCUSSION

Reaching the non-electrified rural population is currently not possible through the extension of the grid, since the connection is neither economically feasible, nor encouraged by the main actors. Further, the increases in oil prices and the unbearable impacts of this energy source on 5the users and on the environment, are slowly removing conventional energy solutions, such as fuel genset based systems, from the rural development agendas. Therefore, infrastructure investments in rural areas have to be approached with cost competitive, reliable and efficient tools in order to provide a sustainable access to electricity and to stimulate development. Renewable energy sources are currently one of the most, if not the only, suitable option to supply electricity in fragmented areas or at certain distances from the grid.

Indeed, renewable are already contributing to the realization of important economic, environmental, and social objectives by the enhancement of security of energy supply, the reduction of Greenhouse gases and other pollutants and by the creation of local employment which leads to the improvement of general social welfare and living conditions. Hybrid systems have proved to be the best option to deliver “high quality” community energy services to rural areas at the lowest economic cost, and with maximum social and environmental benefits. Indeed, by choosing renewable energy, developing countries can stabilize their CO2 emissions while increasing consumption through economic growth.

7.1.1 BATTERY MODE

Initially the circuit is in battery mode, whenever the sunsets, the light around the street light traffic turbines get darken minute by minute. After it the light around the street light reaches certain degrees, the LDR senses and sends signal to the Arduino. As a result, the Arduino automatically turns the relay 1 to connect the battery to the street light led.so the led starts glowing until the energy in the battery is dropped to less than 5 percent.

7.2. TRANSMISSION LINES SUPPLY MODE

Whenever the energy in the battery is low down to less than 5 percent then the relay between the battery and led will be turned off and the relay between the ac to dc converter and led will be turned into closed connection. As a result, the street light led is still goes on, but the power will be saved.

CHAPTER -8

HARDWARE

8.1 OUTPUT



FIG.8.1 HARDWARE

The hardware circuit consists of Arduino, led, ac to dc converter, toggle switch etc as shown in above picture

CHAPTER -9

ADVANTAGES AND APPLICATIONS

9.1 ADVANTAGES

→ there is no need of pointing the turbine in the wind direction.

→ It is efficient because the yaw drive and pitch mechanism are not needed.

→ It will Automatically change the supply whenever it is required to change from Battery to ac supply.

→ There is no need to transfer the Energy generated from the Traffic turbine**.**

9.2. APPLICATIONS

→ Street lights

→ Traffic lights

→ Train electrical applications

CHAPTER - 10

CONCLUSION

10.1. CONCLUSION

Energy plays a crucial role in modern societies. It has a vital input to all sectors like residential, transportation, and manufacture and is essential to generate electricity. In other words, all societies require energy services to meet basic human needs such as lighting, heating, and mobility.

Energy is conserved to reduce consumption costs and to preserve the limited available energy resources. Energy conservation refers to efforts made to reduce energy consumption. The supply of energy on Earth is not infinite. Furthermore, it can take a long time to regenerate energy. By using traffic turbines, we can achieve better energy efficiency and we can reduce energy consumption at high rate.

10.2. FUTURE SCOPE

• The supply change over circuit needs to sense the LDR signal throughout the whole day, The power consumption will be less but not negligible.

• To overcome this situation, we want to activate the supply changing circuit on specific timings like 6 pm to 6 am in our future project implementation.

India ranks fifth in the world in wind power generation at 9600 MW. The coastal region and some parts of Gujarat and Rajasthan in India witness very favourable wind regime, and therefore, the wind power development in these areas has been significant. For commercial exploitation of wind energy, wind velocity at a site should be more than 6 meter per second and corresponding wind power density more than 200 watt per meter sq. In Northern India such high wind velocities are found only on high hilly regions where installation of large-scale wind power projects is itself not feasible due to lack of infrastructure.

CHAPTER -11

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